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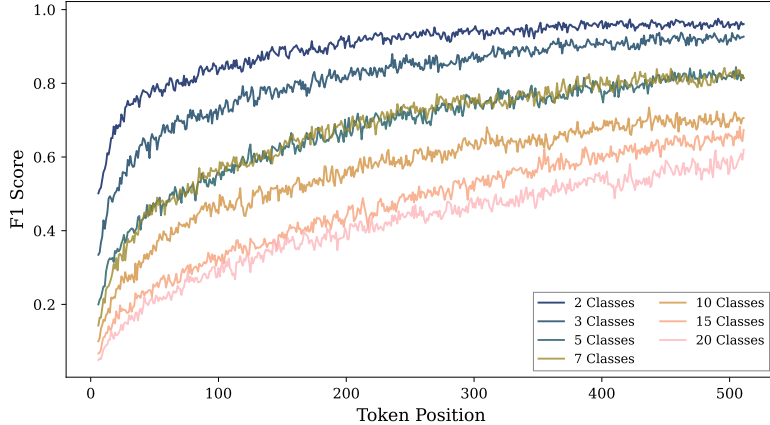


Figure 6. Test F1 score as a function of token position, with attribution tasks involving between 2 and 20 distinctly steered LLMs.

A. Parameterization

This section contains the parameters used to enable reproduction of our results. More details can be found in the code repository (<https://github.com/Thibaud-Ardoin/LLM-Self-Recognition>).

Datasets. The shared repository also contains the text datasets used in our experiments. The custom-curated **Fresh News** dataset was collected via the API of *The Guardian* and consists of clean, plain-text news articles. To minimize potential overlap with model training-data cutoff dates, all articles were published between November 1, 2025 and January 5, 2026. As the dataset is designed for open-ended text completion, we retain only the first six words of each article as the initial prompt provided to the LLM. The **ELI5** dataset was filtered to remove questions containing explicit mention of the word “Edit”, which typically correspond to user-added clarifications appended after the initial question. Questions containing explicit “URL” mentions were also removed, as they often refer to external content that is not accessible to the LLM. For the robustness experiments in Section 3.3, the human reference answer is consistently chosen as the first response listed in the dataset.

Models. Texts are generated with nucleus sampling ($p = 0.9$) and temperature 0.7 for all LLMs and datasets, using the system prompt and query template specified in Appendix D. In the steering scenarios, a repetition penalty is set to 1.1.

Steering. The random sparse steering vectors are sampled from $\mathcal{U}([-1, 1]^d)$, with 99.7% of dimensions set to zero. The scaling coefficient $\alpha = 5$ and the sparsity ratio are selected based on preliminary experiments to balance detection performance and generation quality on Llama-3.1-8B-Instruct. Injection and detection are performed at the middle layer, which empirically yields robust performance across models.

B. Token-wise Analysis

Figure 6 illustrates how the test-time performance varies as a function of token position in the generated sequence. The experimental setup is the MLP classification described in Section 2.3. The earliest tokens are the most difficult to classify, with accuracy close to random chance $\frac{1}{N}$ when distinguishing among N steered LLMs. In contrast, later tokens achieve substantially higher accuracy, approaching the performance obtained by majority voting. While performance could be improved by excluding early tokens from the majority vote, this would break the sequence-length-agnostic nature of the method.

Figure 7 illustrates the cosine similarity between generated text representations and the steering vectors used during generation. With the vector active during steering, we observe an increasing trend similar to that seen in the test accuracy of the MLP setup. For paraphrased texts, the magnitude of this effect is reduced, while the overall trend remains consistent. In contrast, cosine similarity with steering vectors that were not involved in generating the text oscillates around zero or takes negative values.

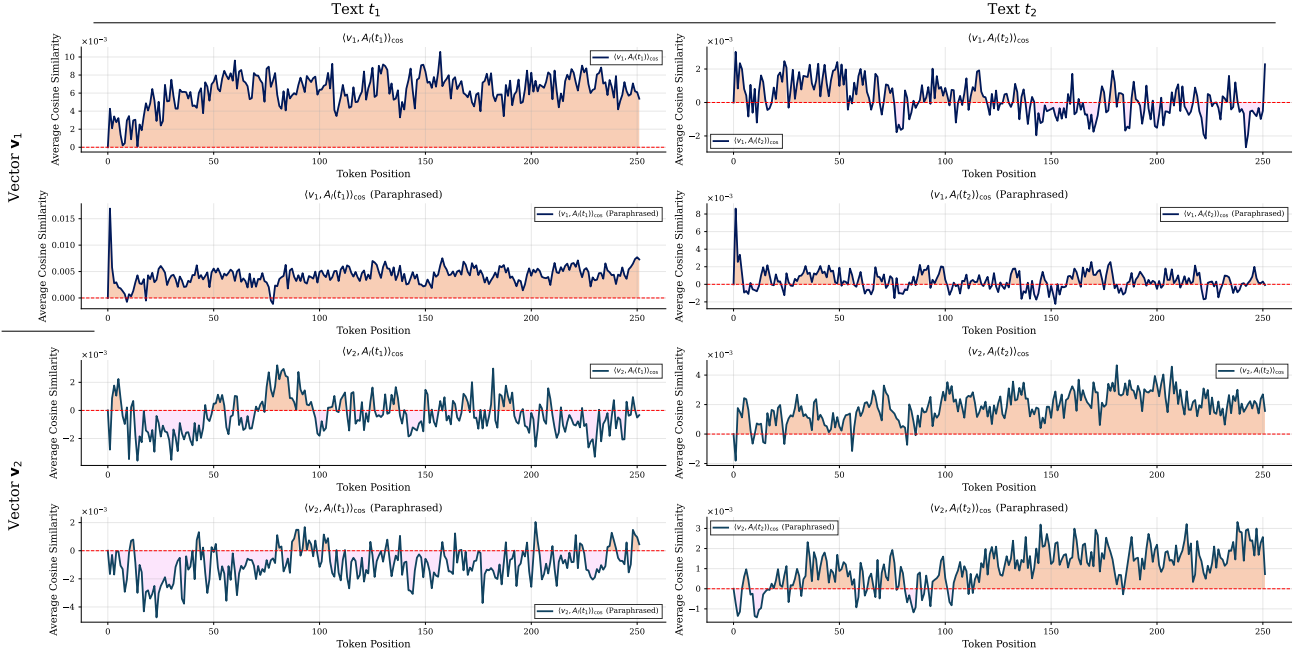


Figure 7. Variation in cosine similarity $\langle \cdot, \cdot \rangle_{\text{cos}}$ as a function of token position. The plots compare the similarity between activations of texts t_1 and t_2 , generated with steering vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, respectively. The paraphrased versions of the texts are also compared.

C. Details of the Paraphrasing Experiments

The paraphrasing model DIPPER-XXL (Krishna et al., 2023) is used with the parameters *Lexical diversity* = 60 and *Reordering* = 20. These parameters are among the strongest setups proposed in the original paper.

Figure 8 presents the results of an alternative robustness experiment to the one in Section 3.3. Because paraphrasing only the watermarked text introduces a distribution shift, we additionally evaluate a setting in which both human- and AI-generated texts are paraphrased. While the KGW watermark is largely unaffected under this new setup, our method exhibits a reduction in accuracy relative to the results shown in Figure 3. This suggests that AI-generated paraphrasing acts as a spoofing mechanism for our representation-based attribution methods.

D. Prompts

Below are the system and user prompts used to evaluate the general self-recognition ability of LLMs (Section 3.1). The prompts were designed to generate summaries that closely match the human-authored ones.

D.1. System Prompt

You write very short summaries of news articles.
 Rules:
 - Output only one sentence whenever possible (but never more than two).
 - Respect the requested word range.
 - The final sentence must end with a single period.
 - No preface, labels, quotes, bullet points, or line breaks.
 - Use a neutral, journalistic tone.
 - Focus on the main event and key entities only.